

RESEARCH REPORT

Market Women Waste Disposal Practices and Disease Risk in Southwestern Nigeria

A survey of 2,000 market women examining how waste disposal practices drive disease risk, and how targeted interventions improved hygiene outcomes across the region.

Prepared by: Samuel Olusesan | May 2026 | Dataset: 2,000 Respondents

2,000

Total Respondents

775

Illness Reported

49.35%

Intervention Coverage

1. Background

Market women in southwestern Nigeria operate in environments where formal waste collection infrastructure is sparse. Many stalls sit within metres of rivers and streams, creating direct pathways for waste to contaminate water sources. The consequences are predictable: waterborne illness, skin infections, and recurring disease cycles that affect both traders and the communities around them.

This study set out to document those patterns precisely, using survey data collected from 2,000 market women across Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti states. The data was cleaned in Microsoft Excel, loaded into MySQL for analysis, and visualised in Power BI.

2. Study Objectives

- Identify the most common waste disposal methods in use across southwestern markets.
- Establish whether a significant link exists between disposal method and reported illness.
- Understand how hygiene knowledge varies across education and income groups.
- Measure the impact of targeted hygiene interventions on disease incidence.

3. Methodology

Structured questionnaires were administered face to face to 2,000 market women across six southwestern states. Responses were recorded across 15 variables covering demographics, waste disposal habits, disease indicators, hygiene practices, and intervention outcomes. After collection, the data was cleaned and standardised in Microsoft Excel, then imported into MySQL Workbench where SQL queries were written to interrogate the dataset. Findings were then brought into Power BI for dashboard visualisation.

Sample Size	2,000 market women
Geographic Scope	Lagos, Ogun, Oyo, Osun, Ondo, Ekiti
Data Collection	Structured questionnaire, face to face
Cleaning Tool	Microsoft Excel
Analysis Tool	MySQL Workbench (SQL)
Visualisation Tool	Power BI
Variables Captured	15 columns across 5 thematic areas

4. Findings

4.1 Illness Prevalence

Out of 2,000 respondents, 775 women (38.75%) reported falling ill in the three months before the survey. Of those, 482 were located within 100 meters of a water body, representing 62.2% of all illness cases. This concentration near water sources points clearly to contamination as a primary exposure route.

4.2 Disposal Methods and Disease Risk

Among women assessed as high disease risk, the two most frequent disposal methods were leaving waste on the ground (172 women) and dumping directly into a river or stream (141 women). Together, these two methods accounted for the majority of high risk cases. Municipal waste bin users represented only 67 of the high risk group, confirming that access to formal disposal options is protective.

Disposal Method	High Risk Count	Illness Rate	Priority
Left on ground	172	High	Immediate
Dumped into river/stream	141	High	Immediate
Buried on site	70	Medium	Monitor
Municipal waste bin	67	Low	None required
Open burning	66	Medium	Short term

4.3 Hygiene Knowledge Across Education Groups

Average hygiene knowledge scores were remarkably close across education levels, ranging from 5.34 for secondary school respondents to 5.52 for those with tertiary education. The narrow gap suggests that formal education alone does not translate into hygiene awareness. This points to the need for targeted, practical hygiene education programmes at the market level regardless of educational background.

4.4 Clean Water Access and Illness

387 women who had no access to clean water at their stalls reported illness in the past three months. This represents 49.9% of all illness cases in the dataset, making lack of clean water one of the strongest predictors of illness in the survey.

4.5 Intervention Outcomes

987 women received a targeted hygiene intervention. Of those, 688 reported a reduction in disease incidence, a success rate of 69.71%. Women who received the intervention scored an average hygiene knowledge score of 5.55 compared to 5.30 among those who did not, a measurable improvement that confirms the interventions were effective beyond just reducing illness.

4.6 Age Group and Disease Risk

Women aged 30 to 44 recorded the highest count of high perceived disease risk at 169 cases, followed by the 18 to 29 age group at 147 cases. This finding is significant because these two groups represent the most economically active segment of the market workforce, meaning disease burden is falling hardest on the women most responsible for household income.

5. Power BI Dashboard Summary

The findings were visualised in a Power BI dashboard branded for EcoVera Waste Solutions. The dashboard surfaces five key views:

- Disease risk level across proximity to water body, showing that women within 50 to 100 metres of a water body recorded the highest concentration of high risk cases (181).
- Waste type per illness reported, with food waste and organic waste accounting for the most illness cases at 166 and 163 respectively.
- Intervention coverage pie chart, showing 1,013 women (49.35%) did not receive an intervention, representing the clearest gap for programme expansion.
- A full respondent table allowing filtering by disposal method, income range, education level, illness type, and access to clean water.

6. Recommendations

- Place covered waste collection points within 100 metres of every market stall located near a river or stream, prioritising the 50 to 100 metre proximity band where high risk concentration is greatest.
- Expand the intervention programme to the 1,013 women who did not receive support. At a 69.71% success rate, this would conservatively result in over 700 additional cases of reduced disease incidence.
- Design hygiene literacy programmes that do not assume formal education as a baseline, given that knowledge scores were nearly identical across all education levels.
- Prioritise the 30 to 44 age group in future intervention rollouts, as they carry both the highest disease risk burden and the greatest economic responsibility within their households.
- Conduct follow-up surveys every six months to track behavior change, monitor disposal method shifts, and measure sustained reductions in illness incidence.

7. Conclusion

The data is unambiguous. Among 2,000 market women surveyed across southwestern Nigeria, improper waste disposal is directly associated with elevated disease risk, and proximity to water bodies compounds that risk considerably. 775 women reported illness in a three-month window. 387 of those had no access to clean water. The two most harmful disposal methods, leaving waste on the ground and dumping into waterways, dominated the high-risk profile.

Targeted interventions worked. Nearly 70% of women who received one reported a drop in disease incidence. The task now is scale. With over 1,000 women yet to receive any form of intervention, and with the market economy depending heavily on the health of the 30 to 44 age group, the case for expanding this programme is clear and well supported by the evidence.

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